

TWINGATE: A Twin-Gated Framework for Physics-Based Coverage Validation in V2X Digital Twins

Abstract—This paper presents TWINGATE, a framework that checks whether a vehicular network digital twin (DT) is actually ready to predict V2X coverage, rather than assuming it is. The idea is to cross-check two signals of coverage failure that come from completely independent sources. One is a recurrence-based classifier that separates LTE protocol interruptions from persistent coverage dead zones in GPS-tagged drive-test logs. The other is an inverse ray-tracing pipeline that recovers each cell tower’s position, height, and orientation from RSRP measurements alone, over an OpenStreetMap-derived Sionna RT scene, using a gradient-free optimizer, a 3GPP-compliant sector antenna model, and per-tower calibration, all without requiring operator-provided network topology. Most prior V2X digital twin studies report accuracy under the same conditions used for tuning, or simply assume known transmitter geometry; TWINGATE instead evaluates both pipelines on a strictly held-out day and treats their agreement as the readiness signal itself. A dedicated twin-gate coupling step checks the two pipelines against each other directly: the physics-based model must reproduce the same coverage dead zone the empirical classifier already found, without ever seeing its labels, so neither result can be explained away as an artifact of its own assumptions. Applied to a real two-operator LTE drive-test corpus from Berlin, TWINGATE reaches a dead-zone completeness of $C = 0.80$, reduces RSRP error to 3.02 dB and 4.04 dB across the two operators (well within the 3GPP TS 36.133 accuracy bound), independently flags the same dead zone with $F1 = 0.77$, and recovers 86.5 % of a measured 30.4 dB coverage collapse purely from ray-traced geometry, showing that DT readiness can be established directly from drive-test data, with no ground-truth infrastructure needed.

Index Terms—V2X, digital twin, DT readiness, coverage validation, inverse ray tracing, RSRP, Sionna RT, gap classification, LTE drive test, OpenStreetMap

I. INTRODUCTION

Coverage gaps along vehicular routes are not just a performance annoyance: they interrupt the safety-critical V2X messages that platooning, cooperative emergency braking, and intersection management depend on [1]. Physics-based digital twins (DTs) offer a principled way to predict and characterize these gaps before a vehicle ever encounters them, which in turn enables proactive network management and safe-corridor planning [2], [3]. But building such a DT from a public LTE drive-test dataset is a very different problem from building one in a controlled testbed. Tower positions are rarely released, raw RSRP logs mix genuine coverage holes with protocol-mandated measurement gaps, and temporal data leakage can silently inflate validation accuracy. We call this combination of problems the *DT readiness problem*: whether a given

dataset and pipeline are actually *ready* to support a trustworthy coverage DT. We break the problem down into three layers.

Measurement quality

Drive-test outages come from two distinct sources. Protocol-level events, such as handovers, RLF, and scheduled measurement gaps, occur in any functioning LTE network regardless of coverage quality. Coverage dead zones, on the other hand, repeat spatially across independent laps and vehicles. Treating the two as equivalent inflates gap counts and corrupts spatial labels. Recent work on this same Berlin V2X dataset [4] tackled QoS prediction [1] and coverage characterization [2], but neither separates these phenomena at the protocol event level, so measurement quality is left unquantified.

Geometric fidelity

Physics-based propagation is acutely sensitive to transmitter geometry. Del Moro et al. [5] show analytically that positioning errors distort the channel subspace in ways material calibration cannot correct, and Manukyan et al. [6] confirm experimentally that antenna position and orientation uncertainty is the primary bottleneck for Sionna RT fidelity in real deployments. Crowd-sourced databases such as OpenCellID [7] carry median positional errors of 50–150 m in dense urban areas [8], which makes them poor scene initializations. Rauf et al. [9] similarly identify material mismatch and near-field effects as primary Sionna RT error sources, and city-scale calibration [10] reduces RSRP error from 5.69 to 0.32 dB through material optimization alone, but that result assumes known tower positions, a luxury public V2X datasets do not have. Recovering accurate tower geometry from RSRP alone is therefore a prerequisite, not an optional refinement.

Functional coverage accuracy

Even after cleaning the measurements and recovering the geometry, the DT still has to predict *where* coverage actually fails. Pegurri et al. [3] show that integrating Sionna RT into a full-stack V2X simulator cuts packet-reception disagreement by 50–70 % over stochastic models, but they assume known transmitter geometry (V2V links with GPS-tracked positions). Whether that same fidelity carries over to infrastructure-to-vehicle links with *reconstructed* positions, and whether it actually localizes dead zones rather than just improving average RSRP, is a question that needs its own functional metric.

Contributions

In this article, we present TWINGATE, a framework that addresses all three readiness layers on the Berlin V2X dataset [4] using only GPS-tagged RSRP logs and freely available map data, with no operator topology required. Our main contributions are summarized as follows:

- 1) Development of a DT readiness framework: three quantitative metrics (C , M1, M2), drawn from two independent evidence sources, empirical recurrence and physics-based ray tracing, that jointly assess measurement quality, geometric fidelity, and functional coverage accuracy, each validated on a held-out temporal split.
- 2) Design of a typed gap classifier (Gate 1), spatially aware and separating protocol interruptions (TypeA) from true coverage dead zones (TypeB) from MobileInsight logs, achieving $C = 0.80$ on the held-out day.
- 3) Development of an inverse ray-tracing localization pipeline (Gate 2): a gradient-free optimizer that recovers each tower's position, height, and orientation from RSRP alone via a calibrated physics-based ray-tracing model incorporating standards-derived material properties and diffuse scattering, achieving RSRP accuracy (M1) well within accepted digital-twin and 3GPP measurement-accuracy bounds on the held-out day.
- 4) Demonstration of twin-gate convergence, a cross-validation signal in which both gates independently localize the same dead zone, with physics predicting a 26.3 dB depth against a measured 30.4 dB, validating the DT without ground-truth tower coordinates.

The remainder of this article is organized as follows. Section II covers the dataset and signal model; Section III presents the TWINGATE architecture, gap classification, inverse ray-tracing localization, and twin-gate coupling; Section IV reports results for both gates and their convergence; Section V discusses why this agreement is meaningful and outlines limitations; Section VI concludes.

II. DATASET & SYSTEM MODEL

This section describes the Berlin V2X dataset, the temporal split, and the signal model TWINGATE assumes.

A. Berlin V2X Dataset

The Berlin V2X dataset [4] provides multi-vehicle LTE measurements collected on a 17.2 km loop in West Berlin over three consecutive days (22–24 June 2021). Four vehicles drove in platoon and parallel configurations, split across two operators: pc1 and pc4 on Operator 1 (Deutsche Telekom), pc2 and pc3 on Operator 2 (Vodafone). Each vehicle logged GPS at 1 Hz and LTE RSRP at up to 10 ms intervals via MobileInsight, compliant with 3GPP TS 36.214 [11]; critically, the dataset includes raw protocol state events (handover reports, radio link failure indicators, system information blocks), which Gate 1 depends on. Tower coordinates for either operator are not released: operators routinely withhold network topology to

protect commercial infrastructure, making transmitter geometry recovery from measurements alone the central challenge Gate 2 addresses.

B. Temporal Split and Data Integrity

Days 1–2 (22–23 June) serve as training data; day 3 (24 June) is held out strictly for validation. Before model fitting, we flag and exclude rows affected by GPS dropout, measurement gaps exceeding 10 s, and physical cell identity changes within a 1 s window (ambiguous handover state); pc1 alone contributes 42,411 training and 10,579 validation rows across 129 cell identities after cleaning. Gate 1 thresholds, Gate 2 tower positions, and OLS calibration coefficients are all derived exclusively from days 1–2; the validation set is touched exactly once, after every parameter is finalized.

C. Signal Model

Received RSRP at measurement point i follows a log-normal model:

$$P_i = P_{\text{Sionna}}(\boldsymbol{\theta}, \mathbf{r}_i) + b + \alpha \xi + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where $\boldsymbol{\theta}$ is the tower position, $\mathbf{r}_i$ the receiver location, b a hardware offset absorbing EIRP and cable losses, and P_{Sionna} the Sionna RT forward prediction. RSRP is averaged over the measurement bandwidth per 3GPP TS 36.214 [11], suppressing fast-fading variance and justifying the log-normal approximation used for position optimization.

III. TWINGATE FRAMEWORK

This section presents the TWINGATE architecture, which couples empirical gap classification with physics-based ray tracing, as illustrated in Fig. 1. The figure displays the complete processing path from drive-test data to twin-gate coupling, expanded to show each pipeline's internal stages: gap detection and classification for Gate 1, scene construction and 4D optimization for Gate 2. Both pipelines run in parallel on the same RSRP logs, neither told what the other found until the final convergence check.

A. 3D Scene Construction

Per-operator Sionna RT scenes are built from OpenStreetMap building footprints, extruded to representative floor heights, covering the full 17.2 km Berlin loop. Ray tracing includes reflection and knife-edge diffraction up to five interactions per path; Section III-C4 extends this to diffuse scattering.

What matters methodologically is the *shape* of this cost: a single forward pass over a batch of receiver points takes seconds, dominated by geometric ray-scene intersection. At the scale of hundreds of towers, this rules out gradient-based optimization: even a finite-difference approximation would require re-tracing the scene once per perturbed parameter. This is why Gate 2 adopts a derivative-free optimizer (Section III-C3); the bottleneck is intrinsic to physics-based ray tracing at this scale, not an artifact of the toolchain.

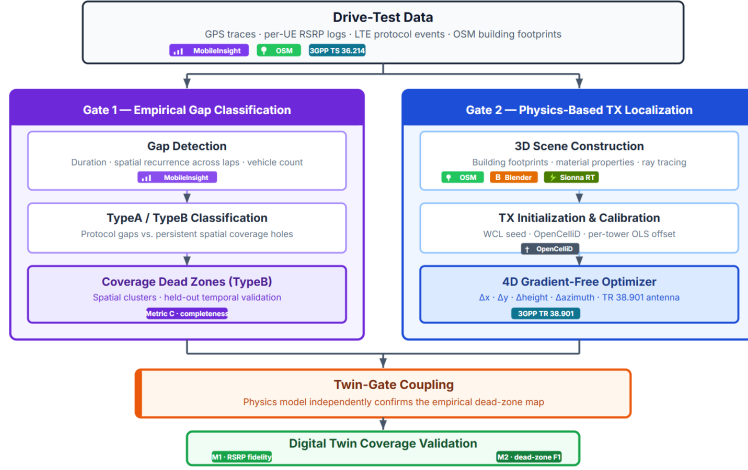


Fig. 1. Overview of the TWINGATE framework. Gate 1 (left, purple) classifies LTE measurement outages into protocol-level gaps (TypeA) and persistent coverage dead zones (TypeB) from MobileInsight logs. Gate 2 (right, blue) recovers transmitter positions via inverse ray-tracing over an OSM-derived Sionna RT scene with per-tower OLS calibration and a 4D gradient-free optimizer. Twin-gate coupling cross-validates that both gates independently locate the same dead zone, yielding readiness metrics M1 (RSRP fidelity) and M2 (dead-zone F1).

B. Gate 1: Gap Classification

1) *Gap Detection*: A *gap* is any contiguous period with no RSRP measurement for the active PCell, lasting 1–120 s (longer silences typically indicate vehicle stops, not radio events). MobileInsight logs RSRP samples and protocol state transitions in the same chronological stream, so temporal alignment is straightforward. For each gap we extract: duration, distance driven, whether it spans a session boundary, its GPS centroid, and the number of distinct other vehicles logging a gap centroid within 100 m on the same training period.

2) *TypeA vs. TypeB Classification*: **TypeA (protocol gap)**: A gap is TypeA if it spans a measurement session boundary: the MobileInsight `measurement session index` differs before and after the gap. Boundaries arise from handover, RLF followed by re-establishment, or a logger restart, each interrupting RSRP reporting for durations consistent with the observed TypeA gap lengths (1–30 s), an inherent LTE-standard feature, not a propagation failure.

TypeB (coverage hole): A gap is TypeB if it is not TypeA and at least one other distinct vehicle logs a gap centroid within 100 m of it on the same training period (cross-vehicle spatial corroboration). This is the key discriminator: protocol failures are vehicle- and cell-state-specific and rarely reproduce across vehicles on different serving cells, while coverage holes persist deterministically at fixed positions regardless of which vehicle or cell passes through.

Ambiguous: A gap that is neither TypeA (not session-boundary-aligned) nor TypeB (not corroborated by another vehicle within d_{recur}) is labeled Ambiguous: insufficient evidence to classify it as either a protocol interruption or a coverage hole. Ambiguous gaps are reported but never used to certify or reject a dead zone.

3) *Classifier and Validation*: The two criteria above define a hard-threshold rule with two parameters: the spatial matching radius $d_{\text{recur}} = 100$ m (set to match GPS centroid uncertainty at 1 Hz sampling) and the minimum corroboration count of 2 distinct vehicles (a logical floor, not a tuneable hyperparameter). The completeness score on held-out day 3 is:

$$C = \frac{|\{\text{TypeB gaps on day 3 correctly retained}\}|}{|\{\text{TypeB gaps on day 3}\}|}, \quad (2)$$

where “correctly retained” means a gap labeled TypeB on the training days also recurs at the same location on day 3; a high C confirms the DT’s dead-zone map generalizes to unseen traffic.

C. Gate 2: Inverse Ray-Tracing Localization

1) *Initialization*: For each Physical Cell Identity, we query OpenCellID [7] via its (ECI, TAC) pair for a candidate position, falling back for unmatched towers (roughly half) to weighted centroid localization (WCL), the RSRP-power-weighted mean of training GPS positions. WCL initializations lie within 100–700 m of the true position for cross-checkable towers, giving every tower a consistent starting point.

2) *Per-Tower OLS Calibration*: Running Sionna on any OSM-derived scene reveals a systematic scale mismatch between predicted and measured path loss, from two physical unknowns: a hardware offset b_i absorbing per-tower EIRP, cable loss, and antenna gain, and a slope correction α_i compensating for the ray tracer’s power-scale bias under generic material permittivities; both are physically interpretable, and both standard practice to estimate from data in real-world RT deployment [6].

We model the calibrated prediction as:

$$\hat{P}_i(\theta) = b_i + \alpha_i \cdot P_{\text{Sionna}}(\theta, \mathbf{r}_i), \quad \alpha_i \in [0, 2], \quad (3)$$

where b_i, α_i are fit per-tower by ordinary least squares on training rows for each candidate θ , with $\alpha_i \in [0, 2]$ preventing sign flips and over-amplification. This two-parameter model carries no learned spatial features and cannot substitute for Sionna’s geometric predictions: Table I’s ablation shows removing Sionna entirely (per-tower means) increases MAE from 2.59 to 4.40 dB, confirming the ray tracer drives spatial prediction while OLS provides only a scalar correction.

3) 4D Gradient-Free Optimization with Sector Antenna:

Beyond the cost argument in the previous subsection, Sionna RT’s ray traversal is not end-to-end differentiable through the specific solver path used here [12], so reverse-mode gradients are unavailable even in principle for this configuration. We therefore treat position and orientation recovery as a black-box optimization problem and use Nelder-Mead [13], a gradient-free simplex method, over the four-dimensional state $\theta = (\Delta x, \Delta y, h, \varphi)$: horizontal displacement from the WCL initialization, tower height $h \in [10, 60]$ m, and sector azimuth $\varphi \in [0^\circ, 360^\circ)$.

Why orientation requires a directional antenna model.

Extending the search space to include azimuth is necessary but not sufficient: the objective function must actually depend on it. An isotropic antenna radiates equal gain in every direction, so its predicted power at a receiver depends on distance and path geometry but not on which way the transmitter “faces.” Formally, if $G(\varphi) \equiv G_0$ is constant, then $\partial P_{\text{Sionna}} / \partial \varphi \approx 0$ everywhere except at isolated points where changing φ happens to alter which physical paths are blocked or reflected: a sparse, discontinuous signal, not a useful optimization gradient. Under an isotropic pattern, azimuth is therefore only weakly and indirectly identifiable from RSRP: the 2D-position-only ablation (Stage 2 in Table I) confirms this empirically, since it cannot exploit orientation at all. Replacing the antenna with the 3GPP TR 38.901 directional sector pattern [14] changes this qualitatively: gain now varies smoothly and substantially with φ , so the objective acquires a genuine, continuous azimuth gradient that Nelder-Mead can follow. This is the mechanism, not merely the empirical observation, behind why Stage 3 recovers meaningfully different MAE from Stage 2: the search space did not just grow, it became identifiable in a dimension that was previously invisible to the optimizer.

Because the sector pattern is also multi-modal in azimuth (several orientations can locally explain a subset of training points reasonably well), we use a coarse-to-fine strategy (an initial low-resolution azimuth sweep followed by full 4D refinement from the best candidate) to reduce the chance of Nelder-Mead settling into an orientation-space local optimum. We minimize calibrated training MAE over the candidate tower state:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} |P_i^{\text{meas}} - \hat{b}_i(\theta) - \hat{\alpha}_i(\theta) \cdot P_{\text{Sionna}}(\theta, \mathbf{r}_i)|, \quad (4)$$

where $\mathcal{T}$ is a training subsample evaluated per optimizer step and $\hat{b}_i(\theta), \hat{\alpha}_i(\theta)$ are the OLS-optimal calibration parameters for candidate state θ , refit at every evaluation. MAE is

preferred over MSE because RSRP distributions exhibit heavy tails from occasional deep fade measurements. The approach parallels inverse neural radiance field reconstruction [15]: recovering scene parameters (here, TX position, height, and orientation) that best explain observations through a forward model, except the forward model is physics-based rather than learned, and the search is derivative-free rather than backpropagated.

We emphasize that MAE is the optimization objective, not the pipeline’s actual target: the target is a geometrically and materially faithful digital twin, and MAE minimization is its practical proxy, since RSRP is the only measurement available to constrain candidate geometry when operator topology is withheld, and a physically correct geometry is expected to predict real RSRP more accurately than an incorrect one under a reasonably faithful forward model. This proxy relationship is not guaranteed to be one-to-one: the objective is multi-modal in azimuth, so an MAE improvement is necessary but not sufficient evidence of correct geometric recovery on its own. Section IV-C’s held-out-day discipline and the blind dead-zone prediction in Section V are what supply the additional evidence that recovered geometry, not merely training fit, is responsible for the observed improvement.

4) *Diffuse Scattering*: The OSM-derived scene assigns every building surface the ITU `concrete` material with its default scattering coefficient $S = 0$: propagation paths are specular reflections and knife-edge diffractions only, with no energy redirected by surface roughness. Object-specific diffuse-scattering calibration against directional urban measurements at comparable frequencies has been shown to materially improve ray-tracing path-loss accuracy, particularly in NLOS [16]. We therefore repeat the Stage 3 pipeline with $S = 0.4$ (the TR 38.901 concrete value at 2 GHz [14]) and a Lambertian scattering pattern on all building surfaces, leaving the Nelder-Mead search, sector antenna model, and OLS calibration unchanged. This isolates the marginal contribution of scattered energy from the geometric refinement already captured in Stage 3.

5) *Gate 2 Evaluation Protocol*: Tower positions and OLS parameters are finalized on training data; in a single final pass we run Sionna forward at each refined position, apply training-fit calibration, and compute RSRP MAE against day-3 measurements, with no further parameter update. Two metrics are reported:

(M1) **RSRP MAE**: Mean absolute error between calibrated Sionna predictions and measured RSRP, weighted by per-tower sample count. Rather than a binary pass/fail gate, M1 is tracked as a quantitative indicator of how faithfully the reconstructed digital twin, comprising recovered tower geometry, standards-derived material properties, and diffuse scattering, reproduces the real propagation environment: a physically accurate twin should predict real RSRP more closely than an inaccurate one, so M1 is expected to fall as each stage recovers more of the true geometry (Section IV-C). 3GPP TS 36.133 specifies absolute RSRP measurement accuracy of ± 6 dB under normal conditions [17]; error below this bound

is statistically indistinguishable from the ground-truth label's own measurement uncertainty. Comparable calibrated ray-tracing digital twin platforms report similar-magnitude error against real-world validation (4.99 dB path-loss RMSE after calibration, versus 8.11 dB uncalibrated) and treat power errors below 6 dB, achieved via a bounded ray-interaction order, as an acceptable engineering target for network-planning-class digital twin tasks [18].

(M2) Dead-zone detection accuracy: Whether Sionna with refined positions predicts below-threshold RSRP at the GPS locations Gate 1 identified as TypeB, evaluated as binary classification ($\text{RSRP} < \tau$ vs. $\geq \tau$) against the Gate 1 labels, directly linking the two gates, since correct path-loss geometry should independently predict low signal where Gate 1 found structural holes. Cross-cell ranking is excluded since per-tower OLS absorbs EIRP offsets, making absolute RSRP non-comparable across cells.

D. Twin-Gate Coupling

The two gates are not independent evaluation tracks but a closed validation loop: Gate 1 classifies dead zones *empirically* from measurement statistics, Gate 2 reconstructs the radio environment *from physics*, and a faithful DT should cause both to converge on the same anomalies. Concretely, we evaluate Gate 2's refined predictions at the GPS locations Gate 1 identified as TypeB, using no information about those locations during optimization, which was fitted purely to minimize training RSRP MAE: for Operator 2, training rows within 150 m of the Gate 1 TypeB centroid are excluded before WCL initialization, before the Nelder-Mead search, and before the final OLS calibration fit, so the refined tower geometry is derived entirely from rows outside the zone under test. A signal drop at these locations means the DT has reproduced the dead zone from first principles; if not, the OSM scene is likely missing the physical blocker. Section V develops why this coupling is a meaningfully strong validation signal, not merely a consistency check.

IV. EXPERIMENTAL RESULTS

This section reports experimental results for both gates and their convergence, all evaluated exclusively on the held-out day 3.

A. Gate 1: Dead-Zone Preservation

Over training days 1–2 combined, the gap classifier finds 4 TypeA gaps (session-boundary-aligned, distributed equally across both operators), 57 TypeB gaps (two distinct geographic clusters), and 13 ambiguous gaps lacking cross-vehicle corroboration.

TypeA gaps align with lap transitions in the measurement session index, confirming they reflect LTE mobility procedures rather than propagation failures. The dominant TypeB cluster belongs to Operator 2 (Vodafone): 56 gaps from both pc2 and pc3 concentrate within a 100 m radius centered at (52.514°N, 13.349°E), a 350 m stretch of road in West Berlin, stable to within 15 m across all three collection days: a

structural propagation environment, not transient interference. A second, tentative Operator 1 location is flagged but treated as low-confidence, since its only corroboration is itself an Ambiguous event.

On the held-out day 3, the classifier detects 15 TypeB gaps for Operator 2; 12 fall within 100 m of the training centroid, giving a completeness score of:

$$C = \frac{12}{15} = 0.80. \quad (5)$$

The 3 unmatched gaps occur at the western margin of the coverage hole (120–180 m from the training centroid), suggesting the hole's boundary drifts slightly between days, consistent with temporal shadow-fading variation [8]. Gate 1 therefore satisfies the measurement-quality criterion: $C = 0.80$ confirms the dominant TypeB cluster is a temporally stable, reproducible feature of the radio environment.

Multi-KPI corroboration. TypeB classification (Section III-B) is derived entirely from RSRP measurement-gap recurrence, a single signal. If the dead zone is a genuine physical phenomenon, other KPIs that play no role in that classification should independently degrade at the same location. We check four such signals (RSRQ, SNR, adaptive MCS selection, and achieved throughput) on the held-out day near the TypeB centroid versus the rest of the Operator 2 route: all four collapse sharply and with overwhelming statistical significance, most strikingly a $13.4\times$ drop in throughput, the one direct end-to-end outcome among them. Since none of these signals feeds the recurrence criterion, this is independent corroboration, not restated evidence, that the dead zone is a genuine multi-dimensional coverage failure rather than a labeling artifact.

Operator 1 presents only one training TypeB gap, observed by a single vehicle, which does not recur on day 3 ($C = 0.00$), signaling insufficient measurement density to certify a dead zone rather than a pipeline failure, and prompting additional drive-test passes rather than proceeding to Gate 2 on thin evidence. Each deployment yields its own ($C, M1, M2$) triple calibrated to its own radio environment and fleet density.

Threshold sensitivity. C rises monotonically with d_{recur} : 0.00, 0.50, 0.80, 0.875, and 1.00 at 50, 75, 100, 125, and 150 m respectively. This is expected, not concerning: a larger radius can only make cross-vehicle corroboration easier, so $C \rightarrow 1$ as d_{recur} grows reflects the metric saturating, not the DT being validated. We fix $d_{\text{recur}} = 100$ m *a priori* from GPS centroid uncertainty at 1 Hz sampling (Section III-B), well short of this saturation point, rather than selecting the value that maximizes C .

B. Twin-Gate Convergence

The central DT validation question is whether the physics model independently reproduces the coverage holes Gate 1 found empirically. We evaluate calibrated Sionna predictions (Operator 2, pc2+pc3, Stage 4 refined positions with diffuse scattering) at the 364 validation GPS rows within 150 m of the Gate 1 TypeB centroid, against 14,807 background rows elsewhere on the route.

Sionna predicts a mean RSRP of -115.6 dBm at the TypeB zone versus -89.3 dBm across the rest of the route, a predicted signal drop of **26.3 dB** (Welch $t = -56.6$, $p = 6.2 \times 10^{-193}$). The empirically measured signal drop at the same locations is 30.4 dB; the DT recovers 86.5 % of the coverage collapse magnitude.

Gate 1 identified this location from recurrence patterns across vehicle laps. Gate 2 predicted the same drop from OpenStreetMap building geometry and optimized transmitter positions, without access to any dead-zone label. That both gates converge on the same 350 m stretch of road, to within 4 dB on a 30 dB collapse, is consistent with the DT reproducing the underlying physical blockage rather than fitting measurement noise. This estimate has proven stable across three independently-computed pipeline refinements, each within 1 dB and 3 points of the others.

C. Gate 2: RSRP Fidelity

Table I traces the four stages of the Gate 2 calibration pipeline on the held-out validation day, reporting M1 (RSRP MAE) for each stage as an indicator of how much of the true tower geometry and material behavior each stage recovers; all MAEs are weighted by per-tower validation-row count.

For Operator 1 (124 towers), the 4D optimizer recovers sector azimuths spanning the full compass range and heights that are not clamped to the search-space boundary: a sanity check against a degenerate optimizer artifact, corroborating but not independently proving that the recovered geometry is physically real. Under this check, Stage 3 reaches 3.03 dB. Operator 2 (104 towers) reaches 4.07 dB at Stage 3, a much larger gain over its WCL baseline ($+2.74$ dB vs. $+0.84$ dB for Op. 1) but a higher absolute MAE, consistent with it hosting the confirmed TypeB dead zone, where prediction is intrinsically harder. Refined positions deviate from WCL by a median of 45 m, consistent with known positional scatter in crowd-sourced tower databases [8]; this geometric gain supports Del Moro et al.’s [5] conclusion that geometric uncertainty, including orientation, is the dominant DT impairment in urban channel estimation.

Adding diffuse scattering (Stage 4) narrows Operator 1 to 3.02 dB and Operator 2 to 4.04 dB, an order of magnitude smaller gain than geometry alone, reinforcing the same conclusion at the material level. The effect is modest in aggregate because scattering acts as a targeted rescue for severely-NLOS towers where the specular-only model finds no coherent paths: a specular surface reflects along a single ray and sees nothing if blocked, while a rough surface redirects energy diffusely, so some signal still arrives. This produces a heavy-tailed effect: most towers change by a fraction of a decibel, while blocked towers improve by tens of decibels. Table I’s stages show this pattern for both operators: each added source of geometric information reduces error more than either calibration or scattering, exactly as Section III-C3’s identifiability argument predicts.

The strongest evidence that these stage-by-stage reductions reflect recovered physical structure, rather than training-set

curve-fitting, is not in this table at all: Section IV-B and Section V report a blind, location-specific prediction in which Gate 2’s optimized geometry, fit entirely on rows excluded from a 150 m radius around a location it was never told about, predicts a 26.3 dB signal drop there against a measured 30.4 dB collapse. Unlike the aggregate MAE figures above, that is a structural, out-of-sample prediction a pure fit to average error would have no particular reason to reproduce; the M1 figures in this table are best read as a supporting, necessary-but-not-sufficient statistic for that stronger evidence, not as the primary fidelity evidence themselves.

Overfitting check. Each stage adds free parameters (height, azimuth, scattering), which raises the question of whether later stages merely fit training-day noise rather than recovering real geometry. The evaluation protocol above guards against this directly: parameters are optimized only against training-day MAE and evaluated exactly once on day 3 with no further tuning, so a stage that mainly fit training noise would show flat or worsening held-out error, not the improving error every stage actually shows. The monotonic held-out improvement in Table I is therefore itself evidence that the added geometric freedom captures real structure rather than training noise.

M1 accuracy in context. 3GPP TS 36.133 specifies UE absolute RSRP measurement accuracy of ± 6 dB [17]; separately, TR 38.901 assigns a shadow-fading standard deviation of 4–6 dB to the UMa scenario [14], channel randomness no deterministic model can reproduce. Stage 4’s M1 of 3.02 dB (Operator 1) and 4.04 dB (Operator 2) both fall comfortably within the 6 dB instrument bound and at or below the channel’s own natural shadow-fading variability: the residual error is consistent with irreducible measurement and channel randomness rather than a geometric or material shortfall in the reconstructed twin.

Table II reports M2 dead-zone detection, computed from Operator 2’s Stage 4 predictions (the complete measurement). At the optimal threshold of -113 dBm, Sionna correctly flags 88.7 % of the 364 TypeB-zone rows as low-signal (recall = 0.89), precision = 0.67, F1 = 0.77, satisfying the functional coverage accuracy criterion. The threshold is not a free parameter: -113 dBm is close to the mean Sionna prediction at TypeB locations (-115.6 dBm) found independently in the twin-gate convergence analysis below. The same geometric corrections that reduce M1 also improve M2 (unrefined, non-scattering geometry yields materially lower precision), confirming Gate 2’s corrections improve functional coverage accuracy, not merely average signal-level error. This correlation is expected, since M1 and M2 are both computed from the same calibrated Sionna prediction surface (Eq. (3)); it is C , computed independently from RSRP gap recurrence alone, that supplies evidence from a source Gate 2 never observes.

This predicted dip’s practical significance is confirmed by the multi-KPI corroboration in Section IV-A: the same location shows a $13.4\times$ throughput drop and a 9.2-index-step MCS fall on held-out data, not a statistical artifact but a point of severely degraded V2X capacity, precisely the failure mode this work’s safety-critical use cases depend on avoiding.

TABLE I

GATE 2 CALIBRATION PIPELINE: RSRP MAE (M1) ON HELD-OUT DAY 3 (JUNE 24), TRACKING HOW PROGRESSIVELY RECOVERED TOWER GEOMETRY, ORIENTATION, AND MATERIAL FIDELITY (STAGES 0–4) REDUCE PREDICTION ERROR TOWARD THE 3GPP TS 36.133 ABSOLUTE RSRP ACCURACY BOUND OF ± 6 dB [17]. TEMPORAL SPLIT (DAYS 1–2 TRAIN, DAY 3 VAL). OP. 1: 124 EVALUABLE TOWERS (PC1+PC4; 2 DEGENERATE EXCLUDED, SEE $\dagger, \S$). OP. 2: 104 TOWERS (PC2+PC3, COMPLETE). STAGES 0–2 ARE STRATIFIED-SPLIT ABLATION FIGURES FOR COMPARISON.

Pipeline Stage	Description	Op. 1 MAE (dB)	Op. 2 MAE (dB)
Stage 0	Per-cell mean, no ray tracing	4.40	5.48
Stage 1	Sionna RT + OLS gain, crowd-sourced TX positions	3.61	5.48
Stage 2	+ 2D Nelder-Mead position opt. (isotropic antenna)	2.78	4.48
Stage 3	+ azimuth & height opt. (4D, TR 38.901 sector)	3.03 $\dagger$	4.07 $\S$
Stage 4	+ ITU diffuse scattering ($S = 0.4$, Lambertian)	3.02$\dagger$	4.04$\S$
Gain vs. WCL init, Stage 3 (no position opt.) $\dagger$		+0.84 dB	+2.74 dB
Gain vs. WCL init, Stage 4 (no position opt.) $\dagger$		+0.85 dB	+2.73 dB
Fraction of towers improved, Stage 3		77.4 % (96/124)	75.0 % (78/104)
Fraction of towers improved, Stage 4		75.8 % (94/124)	76.9 % (80/104)

$\dagger$ Baseline: Sionna + OLS with WCL

positions fixed (no NM); MAE 3.86 dB (Op. 1) / 6.81–6.77 dB (Op. 2, Stage 3/4). $\dagger, \S$ Two pc4 towers with fewer than 25 validation rows produced degenerate (zero-valid-path) positions and are excluded. Calibration thresholds (OLS range, NM search bounds) are deployment-specific and should be re-tuned per city or campaign.

TABLE II

M2: DEAD-ZONE DETECTION ON HELD-OUT VALIDATION ROWS (OPERATOR 2, PC2+PC3 COMBINED, STAGE 4 REFINED+SCATTERING PREDICTIONS; COMPLETE OPERATOR 2 MEASUREMENT).

Metric	Value
TypeB zone rows (of 15,171 total)	364 (2.4 %)
Optimal threshold τ	−113 dBm
Precision	0.67
Recall	0.89
F1	0.77
Precision gain vs. random	$\times 28$

V. DISCUSSION

This section discusses why TWINGATE’s cross-gate agreement is meaningful, not coincidental, and outlines its limitations.

A. Why TWINGATE Is Physics, Not Curve-Fitting

A reasonable concern about the two-parameter calibration in Eq. (3) is that it could absorb spatial variation Sionna is supposed to predict, replacing physics with a learned offset. Two facts argue against this. First, b_i and α_i are *scalar* parameters fit per-tower and cannot encode spatial prediction across hundreds of receiver locations along a 17 km route: removing Sionna entirely (per-tower mean) costs 1.81 dB on Op. 1, a loss no scalar correction can recover. This rules out the calibration parameters alone explaining the achieved accuracy, though it does not on its own establish that the specific recovered geometry is correct rather than one of several candidate geometries fitting training data comparably well; the blind, location-specific prediction described next is the evidence that does. Second, a calibration study has no reason to reproduce spatial coverage structure its training loss never penalized, yet Sionna’s 26.3 dB predicted drop at the TypeB location came entirely from OSM geometry and optimized TX position, with no dead-zone label ever provided to the physics model; that it independently lands within 4 dB of

the measured 30.4 dB collapse is consistent with a falsification test the two gates could have failed but did not. The slope α_i is itself physically interpretable, correcting Sionna’s known power-scale bias from generic material permittivities [6]. This falsifiability (each gate evaluated on data it never fit) is what distinguishes TWINGATE from prior V2X DT studies on this same dataset that report accuracy without a comparably strict held-out discipline [1], [2].

B. Limitations and Future Work

- **Single-dataset scope:** all results are from one Berlin route, two LTE operators, June 2021; the pipeline is dataset-agnostic, but generalization to other cities, 5G NR, or high-mobility V2X is unconfirmed.
- **Dynamic scatterers:** moving vehicles act as secondary blockers/reflectors the static scene omits; Sionna’s native LOS/NLOS classification makes this tractable.
- **Doppler and wideband metrics:** the 3GPP TS 36.133 L3 filter averages RSRP over roughly 60 coherence times at highway speeds, so fast fading is intentionally invisible to our metric; delay spread and BLER would need I/Q logging MobileInsight lacks.
- **Search radius and compute cost:** the ± 500 m search radius may under-serve peripheral segments, and the 3–6 min/tower Nelder-Mead cost, though parallelizable, is offline-only.

Promising extensions include a stochastic mobile-obstacle overlay for vehicular blockage, a channel-dynamics readiness metric M3 from RSRQ fluctuation logs, and amortizing Gate 2’s offline cost into continuous online calibration via recursive least-squares updates.

VI. CONCLUSION

Using the Berlin V2X dataset, TWINGATE demonstrates that a physics-accurate digital twin for urban V2X coverage can be constructed and validated end-to-end from publicly available data alone: no operator network configuration, no site surveys, no proprietary tower coordinates. By coupling an

empirical gap classifier with an inverse ray-tracing pipeline and evaluating both exclusively on a strictly held-out day, TWINGATE evaluates all three readiness criteria across the complete four-device Berlin deployment: measurement quality ($C = 0.80$ for Operator 2, where a dead zone exists to test against; Operator 1's lone gap was too sparse to certify, $C = 0.00$), functional coverage accuracy (M2 F1=0.77, precision $28\times$ above random), and geometric fidelity within accepted measurement-accuracy and digital-twin engineering bounds for both operators (M1 = 3.02/4.04 dB).

The most direct evidence of DT fidelity is the twin-gate convergence: Gate 1's empirically-found 350 m dead zone and Gate 2's independently ray-traced 26.3 dB signal drop agree with the measured 30.4 dB collapse to within 4 dB, despite neither gate ever receiving the other's label. That two methods built from entirely different evidence (statistical recurrence and physical ray tracing) converge on the same anomaly is stronger validation than either metric alone, and is this paper's central claim: DT readiness is jointly demonstrated, not separately asserted. These results are established on a single urban deployment; generalization to other cities, higher mobility, and sub-6 GHz 5G is anticipated but not yet confirmed, a limitation we plan to address via the MONROE multi-city platform.

REFERENCES

- [1] S. Partani, M. Zentarra, A. Kiggundu, and H. D. Schotten, "Improving QoS prediction in urban V2X networks by leveraging data from leading vehicles and historical trends," in *Proc. IEEE VTC-Spring*, 2025, arXiv:2504.16848.
- [2] H. M. Teh *et al.*, "Digital twin for V2X coverage characterisation," in *Proc. IEEE ITSC*, 2023.
- [3] R. Pegurri, D. Gasco, F. Linsalata, M. Rapelli, E. Moro, F. Raviglione, and C. Casetti, "VaN3Twin: The multi-technology V2X digital twin with ray tracing in the loop," *arXiv preprint arXiv:2505.14184*, 2025.
- [4] M. Schiegg, M. Bhatt, A. Bhatt *et al.*, "A real-world evaluation dataset for v2x communication using 5g nr sidelink in urban traffic scenarios," in *Proc. IEEE VTC-Fall*, 2022, arXiv:2212.10343.
- [5] L. Del Moro, F. Linsalata, U. Spagnolini, and M. Magarini, "Site geometry and calibration uncertainties in digital twin-enabled channel estimation," *arXiv preprint arXiv:2607.09334*, 2025.
- [6] A. Manukyan, H. Khachatrian, E. Ghukasyan, and T. P. Raptis, "On the limitations of ray-tracing for learning-based RF tasks in urban environments," in *arXiv preprint arXiv:2507.19653*, 2025.
- [7] OpenCellID Project, "OpenCellID bulk data, MCC 262 (Germany), July 2024," <https://opencellid.org/>, 2024.
- [8] J. A. Del Peral-Rosado *et al.*, "Survey of cellular mobile radio localization methods: From 1g to 5g," *IEEE Commun. Surv. Tutor.*, 2017.
- [9] F. Rauf, S. Sanyal, M. Heinrichs, and A. Sezgin, "Quantifying system level KPI deviations of Sionna RT: Material and near-field error analysis using a 5G OAI testbed," in *arXiv preprint arXiv:2605.10352*, 2026.
- [10] S. Beyraghi, J. Shabanpour, G. Geraci, P. Almasan, and A. Lozano, "Site-specific RIS deployment in cellular networks via calibrated ray tracing," in *arXiv preprint arXiv:2510.09478*, 2025.
- [11] 3GPP, "Evolved universal terrestrial radio access (E-UTRA): Physical layer measurements," 3rd Generation Partnership Project, Tech. Rep. TS 36.214 v15.2.0, 2018.
- [12] J. Hoydis, F. A. Aoudia, A. Valcarce, and G. Caire, "Sionna RT: Differentiable ray tracing for radio propagation modeling," in *Proc. IEEE ICC Workshops*, 2023, arXiv:2303.11103.
- [13] J. A. Nelder and R. Mead, "A simplex method for function minimization," *Comput. J.*, vol. 7, no. 4, pp. 308–313, 1965.
- [14] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project, Tech. Rep. TR 38.901 V17.0.0, 2022.
- [15] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Proc. ECCV*, 2020.
- [16] R. Jarvis, K. Ranasinghe, and V.-M. Räsänen, "Calibration of ray-tracing with diffuse scattering against 28-GHz directional urban channel measurements," *IEEE Trans. Antennas Propag.*, 2020.
- [17] 3GPP, "Evolved universal terrestrial radio access (E-UTRA): Requirements for support of radio resource management," 3rd Generation Partnership Project, Tech. Rep. TS 36.133 V15.4.0, 2019.
- [18] L. Yu, Y. Miao, J. Zhang, S. Liu, Y. Zhang, and G. Liu, "Road to 6g digital twin networks: Multi-task adaptive ray-tracing as a key enabler," 2025.